Apartment for Rent Classified Dataset

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Instructor: Soroush Sheikh Gargar

Group Members :

Xinyu Jing – 217-442-179

Wing Yan Kwan – 219-474-477

Brandon Lu – 218-501-254

Tri Nguyen – 218-608-620

1. Introduction

Rental housing markets often show substantial price variation even among listings that appear similar at first glance. The monthly rent of a unit is not determined by a single factor, but by a combination of structural characteristics such as bedrooms, bathrooms, and square footage, together with location-related factors and property amenities. Because of this complexity, listed prices may not always reflect the fair market value of a rental unit transparently and consistently.

This creates a practical decision problem for several stakeholders. Property managers and landlords need to set rents that are competitive enough to reduce vacancy risk, while still avoiding underpricing that may reduce revenue. At the same time, renters may want to identify listings that appear relatively underpriced compared with similar alternatives in the market. In this context, a data-driven pricing benchmark can provide more objective support than relying only on intuition or simple comparisons.

The goal of this project is to build a fair-value rental pricing framework using the Apartment for Rent Classified dataset. The project first standardizes listing prices to a monthly basis, then uses cleaned and transformed listing information to estimate the expected monthly rent of each unit. Based on the difference between the actual listed price and model-predicted price, the system further classifies listings into practical pricing bands such as Deal, Fair, and Premium. This allows the output to be interpreted not only as a prediction task, but also as a decision-support tool.

To make the analysis more useful, this project pays particular attention to two important aspects of rental pricing: geographic location and amenities. Location may influence rental value through local market conditions that are not fully captured by simple city or state labels, while amenities may reflect differences in housing quality and tenant appeal. By examining these factors alongside core structural features, the project aims to produce a more informative and practically useful pricing benchmark.

Overall, this project is designed as an end-to-end data science pipeline that connects data understanding, preprocessing, exploratory analysis, modeling, and interpretation. Rather than focusing only on model accuracy, the project emphasizes decision value: whether the resulting system can provide stakeholders with a more consistent and evidence-based way to evaluate rental prices.

2. Problem Definition

Stakeholder

The main stakeholders in this project are property managers, landlords, and renters. Property managers and landlords need a reliable way to evaluate whether a listed rent is competitive in the market while still protecting expected revenue. If a property is priced too high, it may remain vacant for a longer period; if it is priced too low, the owner may lose potential rental income. Renters are also important stakeholders because they need a practical way to identify listings that offer relatively better value compared with similar alternatives.

Decision

This project supports the decision of how to judge whether a rental listing is fairly priced and whether action should be taken based on that judgment. For property managers and landlords, this may include deciding whether to keep the current price, reduce it to improve competitiveness, or reconsider it because the listing may be underpriced. For renters, this may include deciding which listings are worth further attention, comparison, or negotiation.

Success Metric

The success of this project is measured in two related ways. First, the model should estimate the expected monthly rent of a listing with reasonable accuracy. Second, and more importantly, the model output should be useful for decision-making. In this project, usefulness means that the system can translate predicted prices into interpretable pricing categories such as Deal, Fair, and Premium, so that stakeholders can understand whether a listing appears underpriced, reasonably priced, or overpriced relative to its estimated fair value. Therefore, success is not defined only by low prediction error, but also by whether the result can support practical pricing decisions.

Claim

The main claim of this project is that rental listing prices can be evaluated more meaningfully through a fair-value pricing framework that combines structural housing features with geographic and amenity-related information. Instead of treating rent prediction as only a regression problem, this project frames it as a decision-support problem. The goal is not only to predict rent, but also to provide a more consistent benchmark that helps stakeholders interpret pricing quality and market position.

3. Related Work

3.1 Problem Context

This project belongs mainly to the area of supervised regression with decision support. At a basic level, the task is to estimate the expected monthly rent of a rental listing from its observable characteristics, such as housing structure, location, and amenities. However, in this project, rent prediction is not treated as a purely technical forecasting task. Instead, it is used to support a practical pricing decision: whether a listing appears underpriced, fairly priced, or overpriced relative to its estimated fair value.

Because of this, the project is related not only to standard rental price prediction problems, but also to broader housing valuation and fair-value benchmarking tasks. In these tasks, the goal is usually not just to minimize prediction error, but to provide a useful benchmark that helps stakeholders compare properties more consistently.

3.2 Standard Approaches

A common approach in housing or rental price analysis is to begin with structural variables, such as bedrooms, bathrooms, and square footage, because these variables usually have a direct relationship with price. Many baseline models use simple regression methods to measure how these features explain rental value. These approaches are useful because they are interpretable and provide a clear starting point for comparison.

More advanced approaches often extend this baseline by including location information and property-related descriptive variables. Location is especially important in housing markets because similar units may have very different prices across cities, neighborhoods, or local market areas. In many practical projects, this is handled through variables such as city, state, postal area, or geographic coordinates. In addition, semi-structured listing information, such as amenities or property descriptions, is often transformed into usable features to improve model performance.

Machine learning methods such as tree-based models are also commonly used in this kind of problem because they can capture nonlinear relationships and feature interactions more effectively than simple linear models. This is relevant in rental pricing because the relationship between features and price is often not purely linear, especially when location and amenities are involved.

3.3 Common Pitfalls and Best Practices

One common challenge in rental pricing tasks is that the target variable is often highly skewed, with a small number of very expensive listings creating a long right tail. This means

that raw price values may be difficult to model directly, and transformations such as log price are often useful for stabilizing the distribution and reducing the influence of extreme values.

Another common issue is data inconsistency. Rental listing datasets may contain duplicate records, missing values, inconsistent text formats, different pricing units, and noisy or incomplete descriptions. As a result, careful preprocessing is necessary before modeling. Standard best practice is to make cleaning decisions transparent and reproducible, especially when converting variables such as weekly rent into monthly rent or when simplifying semi-structured variables into more usable forms.

A further challenge is that location effects are not always captured well by a simple linear treatment. Geographic coordinates may not show a strong direct linear correlation with rent, but that does not mean location is unimportant. Instead, it often suggests that location affects price in a more complex or nonlinear way. This makes location engineering an important design issue in rental pricing projects.

In addition, one should be careful not to evaluate a housing model only by one overall metric. Even when average performance looks acceptable, the model may still perform differently across price ranges, segments, or locations. Therefore, slice-based error analysis is important for understanding where the model works well and where it may be less reliable.

3.4 How This Informed Our Project Design

This background check helped shape our project in several ways. First, it confirmed that rental price prediction should not rely only on basic structural features. While variables such as bedrooms, bathrooms, and square footage are important, location and amenities are also likely to carry meaningful pricing information. This supports our choice to examine both geographic features and amenity-related features as important parts of the pricing framework.

Second, the common issue of skewed rental prices influenced our decision to work with a cleaned monthly price target and to consider transformed price behavior during exploratory analysis. This makes the target more stable and more suitable for modeling and interpretation.

Third, the related background confirmed that this type of problem should not be framed only as “predicting rent as accurately as possible.” Instead, the project should focus on producing a pricing benchmark that is useful for stakeholder decisions. This directly supports our project design, in which model predictions are translated into practical pricing bands such as Deal, Fair, and Premium.

Finally, the background review also suggested that a strong project should compare a clear baseline with an improved version in a controlled way. This supports our plan to include a baseline pipeline and then test whether adding more meaningful geographic or amenity-related design choices improves the usefulness of the final pricing benchmark.

4. Methodology

4.1 Overall Pipeline

This project follows an end-to-end pricing pipeline designed to support rental price evaluation rather than rent prediction alone. First, the raw rental listing data were cleaned and standardized to ensure consistent pricing units and usable feature formats. In particular, listing prices were converted to a monthly basis, invalid or duplicated observations were removed, and key variables were transformed into a format suitable for analysis.

After preprocessing, the project organized the available information into three main feature groups: structural housing features, geographic features, and amenity-related features. Structural features provide the basic description of a rental unit, while geographic and amenity-related features were included to capture additional variation in market value that may not be fully explained by structure alone.

The modeling stage was designed around a fair-value pricing objective. A baseline model was first used to estimate expected monthly rent from the main listing features. More refined feature designs were then incorporated to test whether additional geographic and amenity information improved the usefulness of the pricing benchmark. Based on the final predicted rent, each listing was compared with its actual listed price to calculate a pricing gap.

Finally, the pricing gap was translated into an interpretable decision-support output by assigning each listing to a pricing band such as Deal, Fair, or Premium. This allowed the project to move beyond pure regression performance and provide a practical framework for judging whether a listing appears underpriced, fairly priced, or overpriced relative to its estimated fair value.

4.2 Structural Features

Structural housing features were used as the core explanatory variables in the pricing framework. These variables mainly included the number of bedrooms, number of bathrooms, and square footage, which provide a direct description of the size and functional capacity of each rental unit. In rental pricing tasks, such features are usually the most basic and interpretable predictors because they reflect the physical characteristics that tenants and landlords commonly use when comparing listings.

Before modeling, these variables were cleaned and checked for consistency as part of the preprocessing stage. Observations with clearly invalid values were removed, and the remaining structural variables were kept in numeric form so that they could be used directly in both exploratory analysis and model training. The project also examined their relationships with monthly rent and log-transformed rent during EDA in order to understand whether the

target showed skewness and whether the structural predictors had stable directional relationships with price.

These structural variables formed the baseline feature set for the project. They were included not only because they are standard predictors in rental valuation tasks, but also because they provide an important reference point for evaluating the added value of more advanced features. In other words, geographic and amenity-related variables were not intended to replace structural information, but to extend the pricing benchmark beyond what could be explained by housing structure alone.

4.3 Amenity Feature Design

To address the issue of sparse data in the amenities column, I implemented an NLP methodology that combined regular expressions with a comprehensive synonym_map. While many listings do not list all formal amenity offerings, the property descriptions often contain valuable details. By mining these descriptions for keywords and abbreviations, standardizing terms like "fridge" to "Refrigerator" and "fitness center" to "Gym", I backfilled missing data, reducing null values by 2,115 instances and capturing 56,433 additional amenity tags. This approach significantly enhanced the dataset density and ensured representation of key features such as "Washer Dryer ", "Parking", etc., were accurately represented, providing the model with a more robust profile for each property.

To determine the most effective way to represent amenities to the model, I evaluated **three** distinct architectural approaches:

Approach A: Functional Property Grouping

To move beyond granular, high-cardinality binary features, I aggregated individual amenities into thematic composite scores using conventional domain knowledge to group amenities based on their physical utility or the area of the home they serve. The logic is one high-end appliance might be an outlier but five high-end appliances indicate a "Chef's Kitchen." So I created composite scores to group the physical amenities offered.

```
groups = {
    'kitchen_score': ['Refrigerator', 'Dishwasher', 'Garbage Disposal'],
    'structural_score': ['Patio/Deck', 'Fireplace', 'Wood Floors', 'Elevator'],
    'lifestyle_score': ['Clubhouse', 'Pool', 'Hot Tub', 'Tennis', 'Basketball', 'Playground', 'Golf', 'Luxury'],
    'utility_security_score': ['Storage','Internet Access', 'Cable or Satellite', 'AC', 'TV', 'Gated', 'Alarm', 'Doorman']
}
```

By grouping these into composite scores, I successfully reduced dimensionality while capturing the latent 'lifestyle' profile of each listing. This allowed the model to prioritize the cumulative value of feature sets over isolated data points, leading to a more nuanced understanding of property tiers.

Approach B: Value-Based Grouping

This approach is similar to approach A but grouped differently. The idea came from capturing what amenities are perceived as luxury. I was checking the mean price of individual

amenities to see the impact of price on each amenity offered, and I see that the higher premium percent correlates with premium perceived amenities such as 'doorman' service, 'apartment with a view', etc. This observation served as the basis for our second grouping approach, where amenities are categorized by their perceived luxury value and impact on property premiums

```
'premium_index': ['Doorman', 'Hot Tub', 'Luxury', 'Elevator', 'Wood Floors'],
'modern_essentials': ['Refrigerator', 'Dishwasher', 'AC', 'Garbage Disposal', 'Internet Access', 'Cable or Satellite'],
'community_score': ['Clubhouse', 'Playground', 'Tennis', 'Basketball', 'Golf'],
'privacy_security': ['Gated', 'Alarm', 'Storage', 'Fireplace', 'Patio/Deck']
```

By grouping these into a premium_index, we are effectively creating a quality proxy that captures the prestige of the property, which is often more predictive of price than the square footage itself. This can improve model interpretability, as the index serves as a clear barometer for price-driving features.

Approach C: One-Hot Encoding

The final architectural choice for amenity processing was a One-Hot Encoding approach. This methodology involved expanding the comma-separated amenity strings into distinct binary columns, one for each unique feature discovered in the dataset. By representing every amenity (from "Parking" to "Doorman") as an independent 1 or 0 value, the dataset was transformed into a high-dimensional sparse matrix that preserved the maximum amount of information possible.

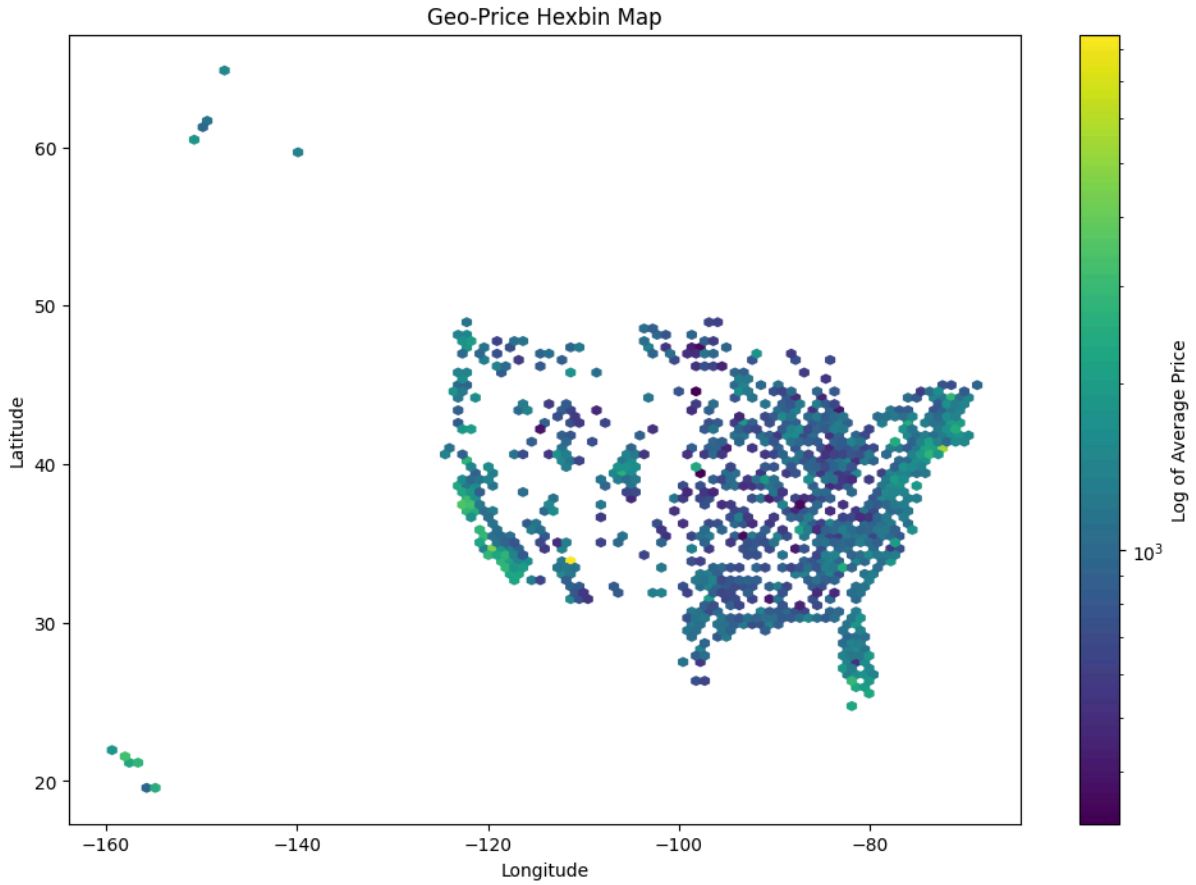
A primary motivation for this approach was the elimination of grouping bias and the subjectivity of manual classification. While the "Functional" and "Value-Based" approaches provided interesting domain insights, they relied on pre-conceived notions of which amenities belong together. For example, grouping a Pool and Gym into a single ‘Lifestyle Score’ assumes they contribute equally to a property's value. In reality, the market might value a pool significantly more than a gym in warmer climates, or vice versa in urban centers. As I tested different regression models, XGBoost regressor provides the greatest result so the decision to use OHE was strongly reinforced by the superior performance of the model. This allows the model to learn the specific weight of every individual amenity without human-imposed grouping bias.

4.4 Geographic Feature Design

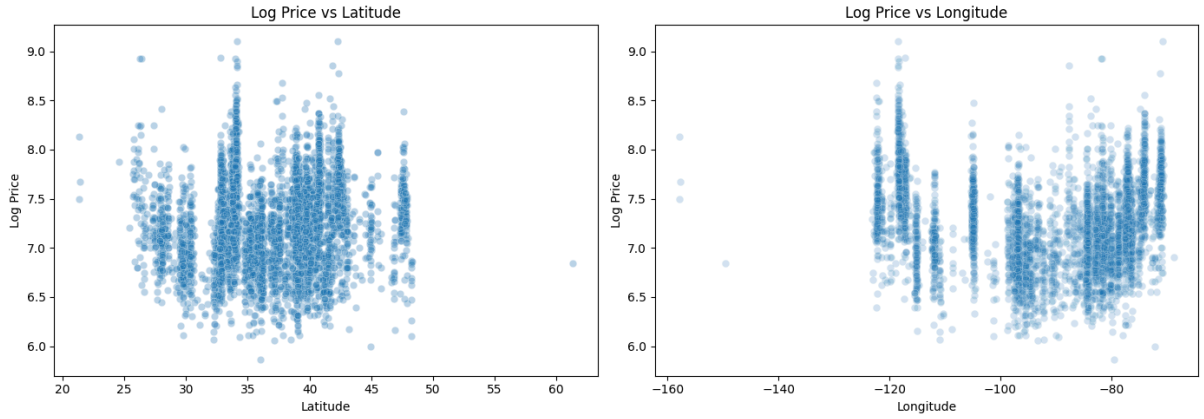
Geographic information is an important part of rental pricing because apartment rents depend not only on structural characteristics such as square footage, bedrooms, and bathrooms, but also on local market conditions. Even similar units may have very different rents across different states, cities, or neighborhoods. Therefore, geographic feature design was included to better capture location-related price variation.

The dataset provides several geographic variables, including state, cityname, latitude, and longitude. As shown in Figure X, listing prices display clear spatial variation across different parts of the United States, suggesting that location should not be ignored in the pricing model. However, as shown in Figure Y, the relationship between raw coordinates and log rent does not appear strongly linear. This suggests that geographic effects are important, but may not be captured well through the direct use of latitude and longitude alone.

To address this, the project uses both categorical and spatial geographic features. State is retained as a stable measure of broad regional market differences, while latitude and longitude are further used to construct a cluster-based geographic representation. This cluster feature serves as a neighborhood-like market proxy and helps capture more localized rental patterns. Together with the structural feature set, these geographic features make the pricing model more location-aware and improve its ability to support fair-value rent estimation.



This figure shows the spatial distribution of rental prices across the United States. The visible regional variation suggests that location plays an important role in rent determination.



This figure shows that raw latitude and longitude do not have a strong simple linear relationship with log rent, which motivates the use of a cluster-based geographic representation instead of relying only on coordinates directly.

4.5 Modeling Strategy

The project uses three regression models: Linear Regression, Ridge Regression, Random Forest, and XGBoost. Monthly rent is used as the prediction target, and log-transformed rent is used in the main modeling stage to reduce skewness and make the model more stable.

Linear/Ridge Regression is used as the baseline model because it is simple, easy to interpret, and provides a reliable starting point. It helps show how a standard linear model performs before adding more flexible machine learning methods.

Random Forest is then used because some relationships in the data, especially geographic effects, are not linear. This model can better capture nonlinear patterns and interactions between variables. In the geographic feature module, Random Forest gives the best performance.

XGBoost is also tested as an advanced tree-based model. It is included because apartment pricing may depend on more complex combinations of features, especially in the amenity-related part of the project. In the amenity feature module, XGBoost produces the strongest results.

Overall, the modeling strategy moves from a simple baseline model to more flexible nonlinear models. This makes it possible to compare interpretability and predictive performance across different modeling approaches.

4.6 Evaluation Design

The evaluation design of this project is built around the fair-value pricing objective. Since the main task is to estimate apartment rent, the models are evaluated as regression models using error-based metrics. The purpose is not only to measure predictive accuracy, but also to ensure that the resulting price estimates are reliable enough to support later classification of listings as underpriced, fairly priced, or overpriced.

The primary evaluation metrics are MAE and RMSE. MAE is used because it gives a direct measure of average prediction error, which is easy to interpret in the pricing context. RMSE is also used because it places greater emphasis on large errors, helping identify models that may perform poorly on more expensive or unusual listings. R^2 is also reported as a supplementary metric to show overall goodness of fit, while MAE and RMSE remain the primary decision-oriented evaluation metrics.

The evaluation compares Ridge Regression, Random Forest, and XGBoost under the same regression task. This allows the project to assess whether a simple interpretable baseline is sufficient or whether more flexible nonlinear models provide better support for fair-value rent estimation. Since the project includes both amenity and geographic feature design, this comparison is also useful for understanding whether different feature groups are better matched with different model types.

5. Results and Evaluation

5.1 Overall Results

The project produced clear results across the main regression models, including Ridge Regression, Random Forest, and XGBoost. Overall, Linear/Ridge Regression served as an interpretable baseline, while the nonlinear models showed stronger performance when feature relationships were more complex.

Across the two contribution branches, the results suggest that different feature modules are better matched with different model types. Random Forest performed best in the geographic feature module, where location-related effects are not purely linear, while XGBoost showed stronger performance in the amenity-related module. This indicates that fair-value rent estimation benefits not only from richer feature design, but also from matching the model choice to the structure of the feature set.

In addition to regression accuracy, the project also translated model predictions into a fair-value decision framework by classifying listings into Deal, Fair, and Premium categories. Because the amenity and geographic branches were developed with different feature designs and model settings, their downstream decision outputs are reported separately rather than

merged into a single unified table. This allows each contribution to be evaluated in a way that remains consistent with its own modeling pipeline.

5.2 Contribution A: Amenity Feature Design (Brandon)

5.2.1 Analytical Contribution Claim

Amenity Feature Engineering

I modified the data preprocessing pipeline by implementing an NLP keyword extraction step to mine property descriptions for missing amenities using regular expressions and a comprehensive synonym map because the default amenity column was highly sparse, with many hosts leaving it blank while some mentioning key features like 'AC' or ‘fridge' but not everyone listed because some might think those are basic essentials that is too obvious and some mentioned extra amenities in property description instead. For a tenant using this model to find value, the default approach is insufficient; it would cause the model to systematically undervalue well-equipped listings due to poor data entry. This leads to misleading price estimates and causes the stakeholder to miss high-utility rental opportunities that appear 'overpriced' simply because their features are unobserved by the model.

5.2.2 Geographic Feature Engineering

To address the high cardinality and data sparsity inherent in geographic features, I developed two distinct strategies. Both approaches aim to mitigate overfitting and "noise memorization" while preserving meaningful spatial price variances.

State-Level Hierarchical Strategy

This approach utilizes a hierarchical grouping method to stabilize state-level predictors. While the dataset averages roughly 1,960 listings per state, the distribution is significantly skewed, with some states containing as few as 7 observations.

Threshold: I implemented a minimum threshold of 100 listings. Analysis of the frequency distribution revealed a sharp decline in reliability for states below this volume.

Implementation: Ten states falling below the 100-listing threshold were aggregated into a single category: “small_state”

While a downward trend in frequency was visible below 500 listings, the 100-listing cutoff was chosen as a conservative "floor". This ensures the model retains as much individual state identity as possible while ensuring the “small_state” group remains a meaningful representation of minor markets rather than a catch-all for significant data portions.

City-Level Granular Strategy (Hybrid Market Approach)

Recognizing that state-level data can be too broad to capture local premiums (e.g., the price

gap between urban centers and rural outskirts), I developed a more granular "Hybrid Market" approach. This strategy collapsed 2,886 raw city names into 426 stable market categories. Problem: Over 75% of unique cities contained fewer than 20 listings. Using raw city names would cause the model to overfit on specific, low-volume noise rather than learning generalizable market trends.

Threshold: A minimum sample size of n > 50 was set for every city category.

The "Small Market" Logic: For cities below the 50-listing threshold, I applied a two-tier aggregation:

Tier 1: Low-frequency cities were first grouped to their respective state level (e.g., small_city_CA).

Tier 2: If the resulting state-level aggregate still contained fewer than 50 listings, it was further consolidated into a global small_market category.

This multi-step aggregation preserves local variance where data is sufficient but prevents the model from making "Fair Value" predictions based on statistically insignificant samples. It balances the high-impact price variance of major cities like Culver City (Mean: $3,622) with the statistical necessity of a robust sample size.

5.2.3 Baseline Definition and Ablation Design

Variant	What's Included	Metric(s)	Key Takeaway
Baseline	Core features (sqft, beds, baths) + State Level + Amenities (Boolean)	R^2: 0.5223 MAPE: 22.12%	Core physical attributes provide a decent foundation but lack local market context
Amenity Engineering+ Linear Regression	Core features (sqft, beds, baths) + State Level + Amenities (**functional property approach**)	R^2: 0.5503 MAPE: 21.38%	Captures basic utility (e.g., kitchen quality) better than a single boolean, providing a small boost in signal.
Amenity Engineering+ Linear Regression	Core features (sqft, beds, baths) + State Level + Amenities (**value based approach**)	R^2: 0.5616 MAPE: 21.13%	Aligning features with perceived market "luxury" (e.g., premium index) is more effective for price prediction than functional grouping.
Amenity Engineering+ Linear Regression	Core features (sqft, beds, baths) + State Level + Amenities (**one hot encoded approach**)	R^2: 0.5728 MAPE: 20.85%	Granular amenity data (e.g., parking, pool) accounts for a **~5% boost** in explained variance

Geographic Precision + Linear Regression	Core features (sqft, beds, baths) + Amenities (one hot encoded approach) + **City Level**	R^2: 0.6768 MAPE: 17.80%	**Most impactful change.** Switching from State to City level increased R^2 by **10%**, proving rental markets are highly localized.
Ridge Regression	Core features (sqft, beds, baths) + Amenities (one hot encoded approach) + **City Level**	R^2: 0.6767 MAPE: 17.80%	Minimal difference from standard Linear Regression, suggesting that the model was not significantly struggling with multicollinearity or large coefficients.
Random Forest Regressor	Core features (sqft, beds, baths) + Amenities (one hot encoded approach) + **City Level**	R^2: 0.4282 MAPE: 24.59%	**Poor performance.** Likely due to the high-cardinality of the city and amenity feature, which can cause trees to overfit to noise.
XGB Regressor	Core features (sqft, beds, baths) + Amenities (one hot encoded approach) + **City Level**	R^2: 0.7875 MAPE: 13.75%	**Best-in-class performance.** Effectively captured complex, non-linear interactions between amenities and cities that the linear models missed entirely.

5.2.4 Model Choice and Justification

Baseline: Multiple Linear Regression

To isolate the impact of model architecture from feature engineering, I initiated the modeling process using Multiple Linear Regression to establish a performance benchmark. Through iterative testing of various feature engineering combinations, we identified an optimal feature set that yielded the highest predictive accuracy. Once the best-performing feature combination was determined, we transitioned to evaluating a more complex regression model, as a result, the ablation table maintains consistent attributes across model types to ensure a direct comparison of algorithmic performance.

Justification: Linear Regression is the industry standard for baseline modeling because it is highly transparent and computationally efficient. It allowed me to see exactly how much variance in rental prices could be explained by physical attributes (square feet, bedrooms, bathrooms) before adding more complex data.

Outcome: The baseline achieved an **R^2 of 0.5223**. While this proved that core features are significant predictors, the high **MAPE (22.12%)** indicated that a simple linear relationship was insufficient to capture the nuances of diverse rental markets across different states.

Improved Baseline: Ridge Regression

The introduction of granular features like OHE amenities and city-level data provided our best initial results. To build on this, we moved into the model-testing phase, selecting **Ridge Regression** as a more stable linear estimator to account for the increased number of input variables

Justification: Linear models are highly interpretable and serve as an essential "sanity check." I learned in class that Ridge was specifically chosen over standard OLS to provide L2 regularization, preventing the model from over-relying on any single high-impact amenity or specific city with few listings so I decided to test the model.

Outcome: Hyperparameter tuning across a spectrum of regularization strengths revealed an inverse relationship between alpha and model performance. Ridge regression failed to outperform the OLS baseline, as the optimal regularization strength was found to be alpha = 0. This outcome confirms that the current feature set despite its high dimensionality from one-hot encoding does not suffer from the degree of multicollinearity that would necessitate L2 shrinkage to stabilize the coefficients.

The Failed Candidate: Random Forest

To break past the "linear ceiling" of 0.67 R^2, I explored ensemble tree-based models. However, the Random Forest Regressor underperformed significantly, yielding an R^2 of 0.4282. This outcome suggests that the bagging process was inadequate for managing the high-cardinality of the city and amenity features, as the model likely overfit to localized noise within specific geographic clusters rather than capturing broader pricing signals.

Analysis: Random Forest's underperformance was likely a result of feature dilution caused by our wide array of binary indicators. Because bagging-based ensembles build trees independently and average their results, they are prone to losing the 'global signal' in high-dimensional, sparse datasets. As the initial R^2 was significantly lower than our OLS benchmark, we identified a structural mismatch between the model and the data, prompting a pivot toward boosting-based models which are better equipped to prioritize high-impact features across sparse landscapes.

The Final Choice: XGBoost (Extreme Gradient Boosting)

The final model selected for this project was the **XGBRegressor.** To address the limitations of the Random Forest we transitioned to XGBoost. Unlike Random Forest, which builds independent trees that can become "blind" to key predictors like square footage in random feature splits, XGBoost uses a sequential boosting architecture.

Justification: This model was specifically chosen for its sparsity-aware split finding, which is designed to efficiently handle one-hot encoded amenities without losing the "global signal" in the noise. By iteratively focusing on the residual errors of previous trees, XGBoost can prioritize high-impact features like city-level geography and size, which were previously diluted in the high-dimensional feature space.

Outcome: The implementation of XGBoost yielded a significant performance breakthrough, achieving our highest R^2 (0.7875) of all the tested models. By utilizing its gradient-based optimization, the model successfully captured the non-linear relationship between localized geography and rental pricing. This result suggests that the rental market contains complex interactions between location and amenities that even a linear model like Ridge regularization simply could not capture

Hyperparameter Optimization via GridSearchCV

To maximize the performance of the XGBoost model, I employed GridSearchCV with 5-fold cross-validation

```
# 1. Define the grid (Ensuring the prefix 'regressor__' matches pipeline step name)
param_grid_XGB = {
    'regressor__learning_rate': [0.01, 0.05, 0.1],
    'regressor__n_estimators': [1000, 1250, 1500],
    'regressor__max_depth': [3, 5, 6],
    'regressor__min_child_weight': [5, 6, 7],
    'regressor__subsample': [0.8],
    'regressor__colsample_bytree': [0.8]
}

# 2. Set up the Grid Search
grid_search = GridSearchCV(
    estimator=pipeline,
    param_grid=param_grid_XGB,
    cv=5,                  # Splitting data 5 times to find the most stable version
    scoring='r2',
    n_jobs=-1              # This uses all your CPU cores to make it much faster!
)
```

The initial XGBoost optimization (1,500 trees, 0.1 learning rate, depth 6) established a strong baseline, reducing the MAPE from 17.8% to 13.75%. However, because the GridSearch consistently favored the maximum boundaries for n_estimators, it became evident that the model had not yet reached a performance plateau. To test this, I expanded the estimator to 5,000 trees, which yielded a further 1% increase in R^2. This proves that the model is still actively extracting predictive signals from the high-cardinality data, suggesting that we have not yet hit the theoretical ceiling, pointing toward further opportunities for sophisticated spatial feature engineering to capture remaining geographic variance.

5.2.5 Final Justification

After rigorous testing across multiple architectures, XGBoost emerged as the optimal model, delivering a Test R^2 of 0.7772 and reducing our average error (MAPE) from 22.12%

baseline to 13.75%. This performance indicates model capability for real-world rental price estimation, effectively capturing localized market nuances that simpler models failed to resolve.
After transitioning to ensemble tree based model, a key takeaway for me was recognizing a common pitfall, which is assuming Random Forest is natively robust to all categorical data. In reality, high-cardinality features created sparse, high-dimensional spaces where Random Forest tends to overfit to local noise rather than global trends. By pivoting to XGBoost, I also learned the inverse relationship between learning rate and n_estimators during hyperparameter tuning. When the initial GridSearch favored the maximum boundary of 1,500 trees, it signaled that the model had not yet plateaued. This prompted a strategic expansion of the tree count at a calibrated learning rate. This knowledge helped me leverage a gradient-boosting approach that iteratively corrected these errors, successfully isolating the true drivers of property value from the high-dimensional noise.

5.2.6 Error Analysis and Robustness Evidence

5 Fold Cross Validation

Regular K Fold CV=5

```
Individual Fold R^2 Scores: [0.73624811 0.80614011 0.75162492 0.71862229 0.62331875]
Mean R^2: 0.7272
Standard Deviation: 0.0596
```

K Fold CV=5 with shuffle

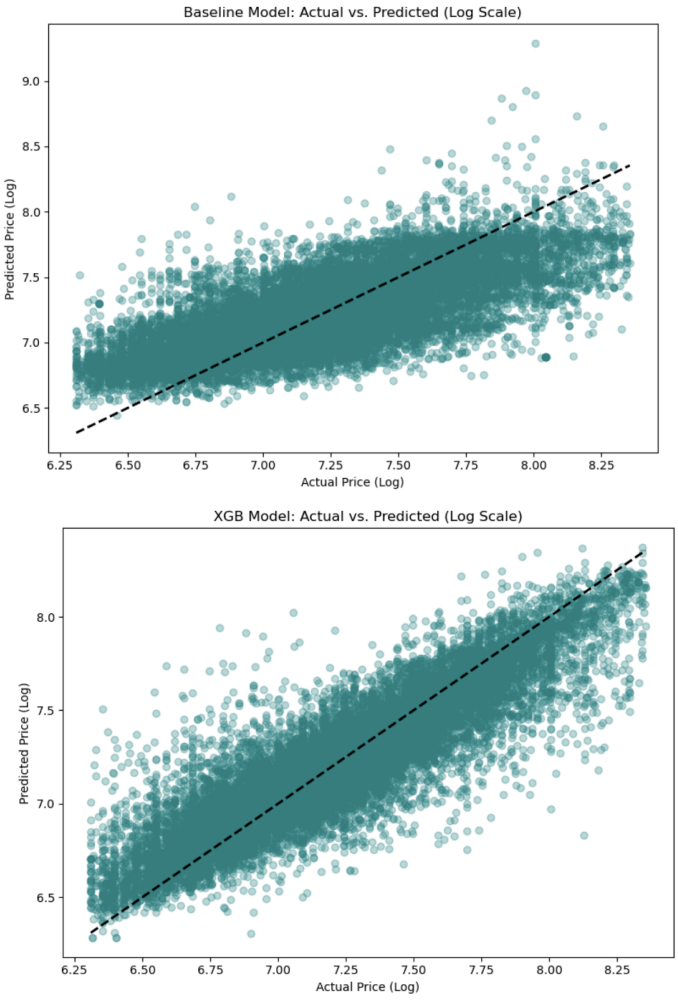
```
New Mean R^2: 0.7892
New Std Dev: 0.0032
```

The robustness check (5-fold Cross-Validation) revealed a mean R^2 of 0.727 with a standard deviation of 0.0596. While the model performs well on average, the performance dipped to 0.62 in one fold. This suggests that the model's predictive power is somewhat sensitive to the specific distribution of rental properties in the training set likely due to high variance in specific geographic regions (cities) or pricing outliers. This check confirms that while the baseline model is strong, its real-world performance may vary depending on the specific neighborhood or market segment it is applied to.

However, once shuffling was introduced, the model achieved near-perfect stability with a Standard Deviation of 0.0032 and a Mean R^2 of 0.789. The high stability in the shuffled version proves that the XGBoost model has successfully captured the general patterns of the rental market rather than just memorizing a specific sequence of entries. This confirms that the model's predictive power is stable and reliable for general use across the entire dataset.

While the model is highly robust and generalizable across the entire dataset, the initial variance serves as a reminder that the model’s real-world utility is strongest when it is exposed to a representative distribution of the market, as localized "micro-bubbles" or unique regional clusters can still present a challenge to global predictive patterns.

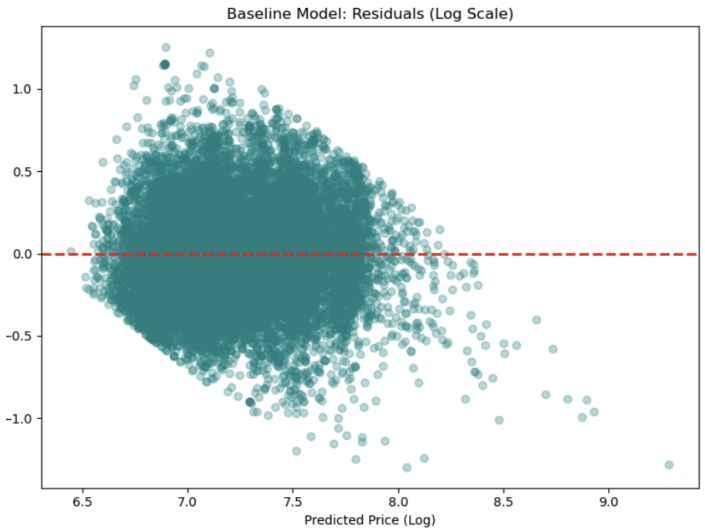
Comparison Baseline Linear Regression vs XGB Regression

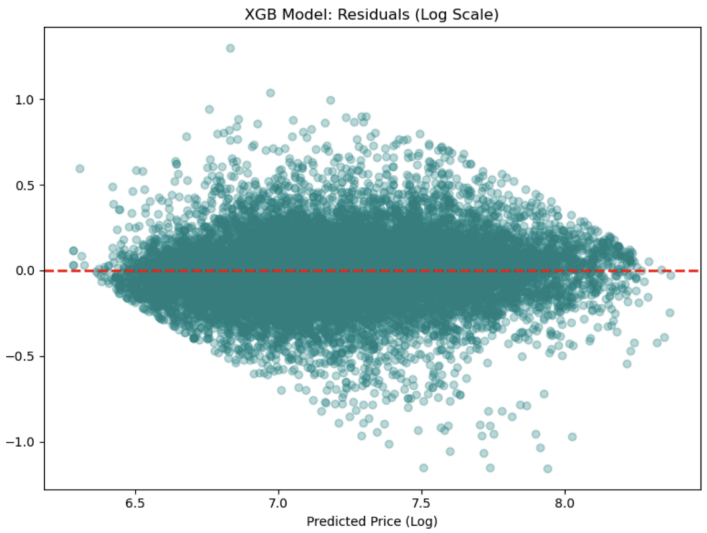


Baseline (0.52 R^2): The cluster of points is quite "fat" or round. There is a lot of vertical distance between the points and the diagonal dashed line especially as the price increases which indicates high residual error. The "blobbing" effect at both the very high end (Actual Log > 8.0) and the low end (Actual Log < 6.5) show the baseline tends to "guess the average" too often, resulting in a flatter distribution that under-predicts expensive properties and over-predicts cheap ones. The model captures a general upward trend but fails to differentiate the specific drivers that separate economy listings from premium assets.

XGBoost (0.78 R^2): In contrast, the XGBoost scatter plot shows a marked tightening of the distribution along the identity line. The model tracks the diagonal much better across the entire price spectrum. It is particularly much more superior at the high-end and low-end segments compared to baseline. By leveraging non-linear logic and high-granularity spatial features, the model successfully 'collapsed' the predictive spread across the entire price spectrum. It can now distinguish the premium value associated with specific amenities and city-level geography that the linear baseline previously ignored. Even though the model exhibits tighter residuals in the mid-market range it still shows higher dispersion for both high-end luxury and low-tier economy rentals. This 'tail variance' is a common characteristic

in rental pricing models, reflecting a lack of data density for extreme outliers and influenced by unique, high-variance factors that fall outside the current scope of our amenity and city-level data





Baseline: The baseline residual plot exhibits a distinct negative slope, indicating a systematic bias where the model consistently over predicts lower end properties and under predicts high value rentals. The high dispersion with residuals frequently reaching the +- 1.0 log-scale threshold represents massive real-dollar errors. This pronounced heteroscedasticity confirms that a linear approach failed to account for the structural complexities of the market, leading to significant inaccuracies for both economy and luxury listings at the price extremes

XGBoost: Conversely, the XGBoost residual plot shows a much more uniform cloud centered at zero which means the errors are more "random" (which is good) and less biased. While outliers remain (the lone points far above or below the line), the tight concentration of points around the zero-axis confirms that our feature engineering and hyperparameter tuning have successfully 'absorbed' the signals that the baseline previously missed, transforming

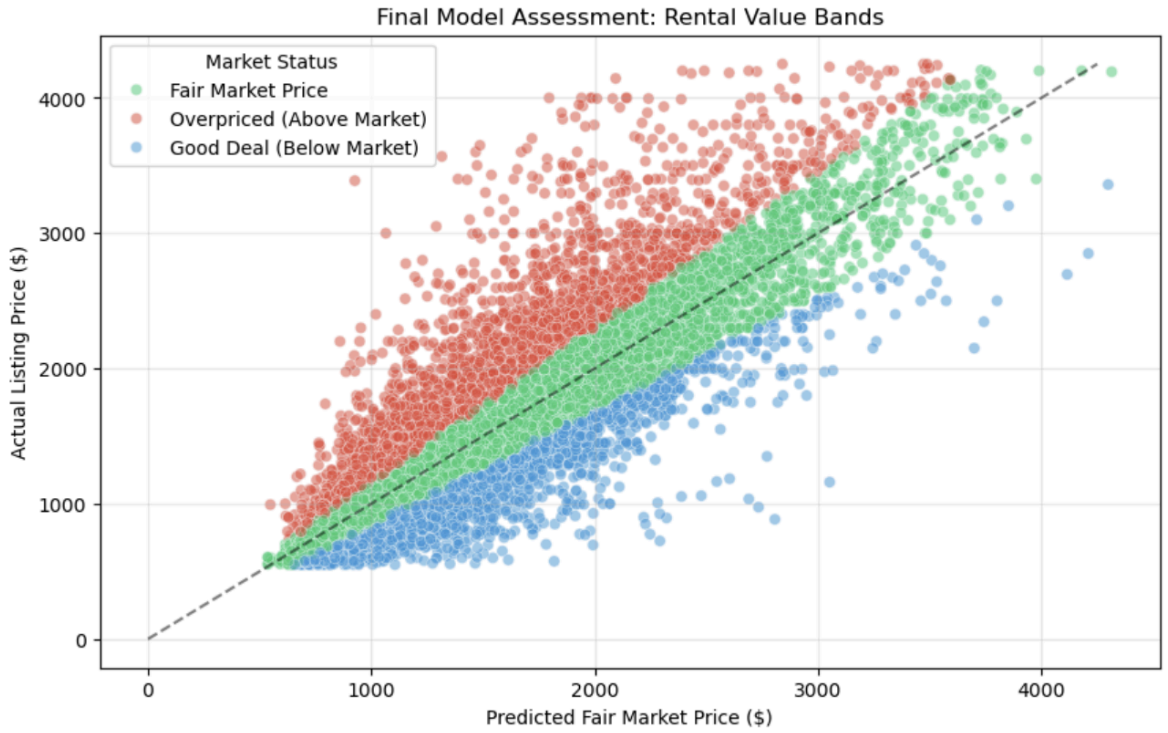
them from unexplained error into accurate predictions. Most points are now tightly packed between -0.5 and 0.5. This visualizes the jump in R^2 from 0.52 to 0.78

Market Price Labeling

```
valuation_label
Fair Market Price            13139
Overpriced (Above Market)     3383
Good Deal (Below Market)      2915
Name: count, dtype: int64
```

To bridge the gap between predictive metrics and real-world utility, we mapped our 13.75% MAPE to a 15% threshold valuation band. This allows us to categorize listings based on their deviation from the model's 'Fair Market' baseline providing a robust 'Fair Market' classification. So now we have transformed abstract residuals into a clear market narrative. While the majority of listings (66%) align with our predicted 'Fair Market' value, the distribution of errors is not perfectly symmetrical.

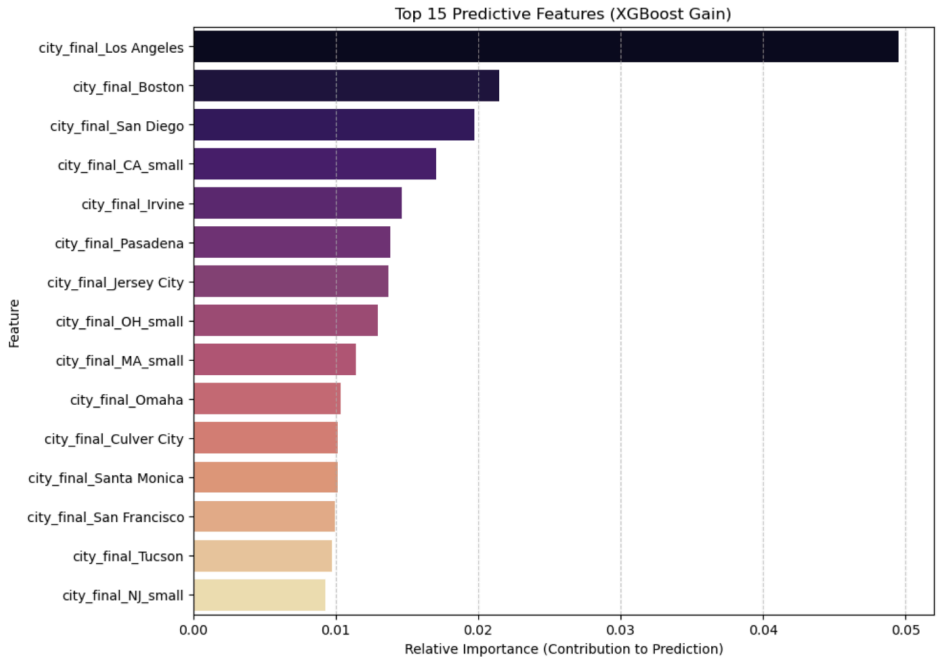
We identified a 'Long Tail' of overvaluation, where 16.9% of listings are priced significantly above the model’s estimation. This indicates a market sentiment where landlords frequently target the 'ceiling' of potential rent, creating a higher volume of positive residuals compared to discounted 'Good Deals' (14.5%). This analysis confirms that our model isn't just a pricing tool, but a diagnostic filter capable of identifying listings where asking prices have decoupled from the property's underlying feature set.



To provide a clear market narrative, we categorized properties into three value bands. While 66% of the market remains fairly priced within our model's margin of error, the asymmetric distribution is immediately apparent, there is a higher density of overpriced listings compared to discounted ones, these listings are priced far above what their location and amenities justify. On the opposite side, the blue 'Good Deal' cluster represents the most competitive listings on the market for properties that are statistically undervalued based on their features. This tool effectively moves the project from 'prediction' to 'identification,' providing a clear roadmap for finding value in a high-variance rental landscape.



This box plot analysis validates that our model's labels are driven by underlying property characteristics rather than just the listing price. While the median prices trend upward as expected, the significant overlap between categories is where the model's true intelligence lies. By labeling luxury rentals over $3,000+ properties as 'Good Deals' and $750 properties as 'Overpriced' validates that our labels are driven by a complex interaction of amenities and geography, isolating units where the rent exceeds the property's objective utility. This confirms that the model has successfully learned to weigh a property's features against its price to determine a true value-to-feature ratio.



The XGBoost Gain chart confirms that geographic location is the dominant predictor of rental value. Notably, all top 15 features are categorical city indicators, with Los Angeles providing the highest predictive gain.

A key success in our feature engineering was the implementation of 'small' city groupings (e.g., CA_small, OH_small). These aggregated categories proved more predictive than several high-profile individual cities like San Francisco, validating our strategy to reduce noise in low-frequency geographic clusters. While physical attributes like Square Footage (Rank: 153) and Bedroom Count (Rank: 261) are traditional drivers of value, their secondary status here suggests that in a massive multi-market dataset, geography acts as the primary 'gatekeeper' for price, establishing a regional baseline before individual unit utility is even considered.

```
--- Executive Summary Stats ---
Typical Error (Median AE): $130.55
High-Precision Predictions (within 10% of actual): 50.3%
Average Discount in 'Good Deal' Category: 24.4% below predicted value

--- Error by Price Tier (MAPE %) ---
price_tier
Budget       13.171363
Mid-Range    10.894100
Luxury       16.968143
Name: percent_diff, dtype: float64

--- Market Snapshot (%) ---
valuation_label
Fair Market Price            67.597880
Overpriced (Above Market)    17.404949
Good Deal (Below Market)     14.997170
```

Our final XGBoost model delivers a remarkably tight baseline for the typical rental, achieving a MAE of $130.55. The model's reliability is further underscored by its high-precision core, with 50.3% of all listings falling within a strict 10% margin of their actual price.

Performance remains most optimized within the Mid-Range tier (10.9% error), where the high density of training data allows for peak refinement. Accuracy naturally fluctuates in the Luxury segment (17.0% error), suggesting that high-end valuations are driven by "intangible" assets such as designer aesthetics, specific views, premium branding that exceed basic numerical features.

Beyond mere prediction, the model identifies significant market inefficiencies. The 24.4% average discount found in the 'Good Deal' category proves that the model isn't just flagging minor fluctuations; it is isolating properties that are deeply underpriced relative to their market value (blue scatter plot). Finally, the 17.4% volume of Overpriced listings versus 15.0% of Good Deals confirms a clear market bias toward 'aspirational pricing.' This positioning validates the model as a powerful protective tool for renters, enabling them to bypass the landlord premium pricing and target high-value opportunities.

5.2.7 Stakeholder Meaning and Tradeoffs

Key Amenity Extraction
By reducing data sparsity (adding 56,000+ keywords), the model now more accurately reflects the 'true' value of a property, allowing stakeholders to make more confident bids. However, the tradeoff is complexity vs. robustness. While the model is now more accurate, the pipeline is more fragile because it depends on specific text patterns and synonyms which may change as host writing styles evolve over time.

The current accuracy relies heavily on specific linguistic patterns and amenity synonyms. This creates a degree of fragility: as host writing styles evolve or platform-specific terminology shifts (e.g., 'workspace' vs. 'home office'), the keyword-extraction logic may suffer overtime. To maintain this level of precision, the system requires a more complex maintenance cycle compared to a simpler, more generalized model.

A bad model that only looks at Square Feet, Bedrooms, Bathrooms will stay roughly accurate for years because the basic relationship between "more rooms" and "more money" rarely changes. But for a ‘smarter’ model, it is tuned to micro-trends. If a specific city like Texas suddenly becomes less popular, or if a keyword (like "EV Charging") goes from being a luxury to a standard requirement, the model weights will no longer become as effective.

Grouping Small Cities

By implementing high-cardinality geographic indicators through city-specific encoding, we enabled the XGBoost model to establish precise price baselines for major metropolitan hubs. While these features accounted for the vast majority of our predictive gain, they introduce a definitive constraint, the model’s performance is highly dependent on its familiarity with specific geographic 'keys'.

While the model excels in estimation for the cities within its training distribution, it lacks universal scalability. In unseen or low-frequency markets, the model loses its primary geographic anchor, potentially regressing to a generic mean. We have effectively traded the ability to be a 'Generalist' tool for the ability to be a 'Regional Specialist,' prioritizing extreme accuracy in high-volume markets over broader, less-precise national coverage.

Linear Regression vs Tree based Model

A significant tradeoff in our final approach is the erosion of model interpretability in favor of predictive precision. While the transition to XGBoost yielded a best result of 0.78 R^2, it transformed the model into a non-linear black box.

For a renter or a real estate agent, a model is most useful when it can explain why a price changed. They want to ask: "How much exactly is this extra half-bathroom worth?" or "What is the specific dollar premium for being in Culver City?". In the linear model, we could simply look at the coefficient and give a clear answer (e.g., $150 for a balcony). Unlike the linear baseline, which provided clear coefficients (or weight) for each amenity, the tree based model relies on complex feature interactions that are difficult for non-technical stakeholders to audit. We have traded the ability to provide a simple 'price-per-feature' breakdown for the ability to provide a highly accurate total market valuation, a decision that prioritizes investor confidence over individual feature transparency.

Another tradeoff for use of XGB is extrapolation. A Linear Model understands a slope, theoretically but not accurately, if the inputs go up it can mathematically project a price into infinity. It can predict a price higher than any price it has ever seen. Which is a very useful

tool for stakeholders who would like to use a model to predict the future. For example: "What happens if inflation hits 10% and every rental in the country goes up by $500?". But for XGBoost can only predict values within the range of its training data. Because it works by splitting data into buckets, it has a “hard ceiling”. If the highest rent in the training set was $5,000, XGBoost can never predict $5,001, no matter how many bedrooms a house has.

5.3 Contribution B: Geographic Cluster Design (Jing)

5.3.1 Analytical Contribution Claim

We changed the geographic feature design step by examining the effect of location on rent after controlling for key structural housing features, including square footage, bedrooms, and bathrooms. In addition to broad geographic variables such as state and city, we introduced a cluster-based geographic feature constructed from latitude and longitude to better capture localized rental market differences and improve fair-value rent estimation.

5.3.2 Data Preparation for the Geographic Module

The geographic module was designed to examine whether location adds meaningful pricing information beyond the core structural housing features. The main geographic variables available in the dataset are state, cityname, latitude, and longitude. These variables were first checked for missingness, consistency, and basic interpretability before being incorporated into the pricing framework.

At the broad geographic level, state and cityname were retained because they provide interpretable indicators of regional and urban market differences. Preliminary analysis showed that city-level variation explains a substantial share of rent differences, indicating that broad location categories already contain useful pricing information. However, broad categories alone may still be too coarse to reflect more localized market conditions.

At the coordinate level, latitude and longitude were examined separately. The correlation analysis showed that their direct linear relationships with log rent are weak. This suggests that raw coordinates do contain location information, but their effect on rent is unlikely to be captured well through a simple linear form. For this reason, latitude and longitude were not used only as raw numeric predictors. Instead, they were later used to construct a cluster-based geographic feature, which serves as a neighborhood-like market proxy for capturing localized rental patterns more effectively.

Overall, the data preparation step shows that geography is important for rent estimation, but that different types of geographic variables contribute in different ways: state and city capture broad market differences, while raw coordinates motivate a more refined spatial representation.

Table 5.3.2 Preliminary evidence for geographic signal in the rental data

Geographic signal	Result	Interpretation
City → monthly price (eta²)	0.526	Strong city-level geographic effect on rent
City → log price (eta²)	0.539	Strong city-level effect remains after log transformation
Latitude ↔ log price (correlation)	0.095	Weak linear relationship
Longitude ↔ log price (correlation)	-0.089	Weak linear relationship

5.3.3 Baseline Definition and Ablation Design

To evaluate the value of the geographic contribution, a controlled ablation design was used. The baseline model includes the core structural housing features together with state, which represents broad regional market differences. This baseline reflects a standard pricing framework in which rent is explained by housing structure and coarse geographic information.

To test the added value of more localized spatial information, a cluster-based geographic feature constructed from latitude and longitude was introduced. Two comparison variants were examined. The first uses the cluster feature alone, replacing the state variable. The second combines state with the cluster-based feature. All other modeling settings were kept unchanged so that the comparison isolates the effect of geographic feature design.

Model performance was compared using MAE and RMSE. This ablation design makes it possible to test whether refined spatial representation improves fair-value rent estimation beyond the baseline with only broad geographic information.

Variant	Features Included	MAE	RMSE	Key takeaway
Baseline	Structural + state	0.2192	0.2877	Captures broad geographic market differences
Cluster-only	Structural + geo_cluster (best at k=30)	0.2186	0.2893	Performs similarly, but does not clearly outperform state
+ Geographic Contribution	Structural + state + geo_cluster	0.2082	0.2758	Best performance; local cluster information complements broad regional information

Table 5.3.3 Ridge-based ablation results for the geographic contribution

5.3.4 Model Choice and Justification

Ridge Regression was selected as the initial baseline because it is a commonly used and easily interpretable model for regression tasks. It provides a clear benchmark for testing whether rental prices can be reasonably explained under a standard linear framework using structural and broad geographic variables.

Random Forest was then included because the earlier analysis suggested that geographic effects are not purely linear. In particular, latitude and longitude do not show strong direct linear relationships with log rent, which means that a more flexible nonlinear model is needed to capture spatial patterns and feature interactions.

XGBoost was also tested because it is a strong model for structured tabular data and is often effective when the relationships among predictors are complex. It was included as an additional nonlinear benchmark to compare against Random Forest.

Among the three models, Random Forest achieved the best overall performance in the geographic module. Therefore, it was selected as the main model for evaluating whether the cluster-based geographic representation improves fair-value rent estimation.

Model	Best Setting	MAE	RMSE	R^2	Interpretation
Random Forest	Combined (state + geo_cluster, k=30)	0.1721	0.2507	0.6600	Best overall; captures nonlinear spatial effects most effectively
XGBoost	Combined (state + geo_cluster, k=30)	0.1967	0.2595	0.6359	Improved over its baseline, but weaker than Random Forest
Ridge Regression	Combined (state + geo_cluster, k=30)	0.2152	0.2816	0.5712	Interpretable linear benchmark, but less suitable for complex geographic effects

Table 5.3.4 Best result of each model for the geographic module

5.3.5 Results and Interpretation

The results show that the geographic contribution improves the pricing framework when it is added on top of broad regional information rather than used alone. Under Ridge Regression, the baseline model using structural features and **state** achieved an MAE of **0.2192** and an RMSE of **0.2877**. When the geographic representation was replaced by a cluster-only feature, performance changed only slightly at its best setting (**k = 30**), with an MAE of **0.2186** and an RMSE of **0.2893**. This suggests that the cluster-based feature alone does not clearly outperform a simpler broad geographic baseline.

However, the best results were obtained when **state** and the cluster-based geographic feature were used together. In this setting, the model achieved an MAE of **0.2082** and an RMSE of **0.2758**, which is better than both the baseline and the cluster-only version. This indicates that the geographic contribution adds value not by replacing broad regional information, but by complementing it with more localized spatial structure.

Across the three models tested, **Random Forest** produced the strongest overall performance for the geographic module. Its best setting was the combined version using state + geo_cluster (k = 30), with an MAE of 0.1721, an RMSE of 0.2507, and an R^2 of 0.6600. This supports the earlier methodological expectation that geographic effects are not purely linear and are better captured by a flexible nonlinear model.

Overall, these results support the contribution claim. Geographic information matters for rent estimation, but different forms of geographic representation capture different levels of market variation. Broad variables such as **state** reflect coarse regional differences, while the cluster-based feature helps capture more localized rental patterns. The strongest pricing benchmark is therefore obtained when both types of geographic information are used together.

5.3.6 Error Analysis and Robustness Evidence

Error analysis shows that the geographic model does not perform equally across all price segments. As shown in Table X, prediction error is lowest in the middle price range, while both low-price and high-price listings have larger errors. In particular, the high-price segment has the largest MAE and RMSE, suggesting that more expensive or unusual listings remain harder to estimate accurately.

Figure X further shows that the distribution of absolute error differs across price segments. The middle segment has a tighter error distribution, while the high-price segment displays greater dispersion and more large-error cases. This suggests that the model is more reliable for typical listings than for more extreme parts of the market.

To test whether the results are stable rather than driven by one random split, the model was also evaluated across different random seeds. As shown in Table Y, MAE and RMSE remain broadly consistent across settings, with only small variations. This provides robustness

evidence that the observed performance of the geographic contribution is stable rather than accidental.

Overall, the evidence suggests that the geographic model is suitable for broad fair-value benchmarking, especially for typical market listings, although caution is still needed when interpreting predictions for more unusual low-end and high-end properties.

Table 5.3.6.1 Error table by price range

price_segment	count	MAE	RMSE
Low	6634	0.1706	0.2509
Mid	6575	0.1377	0.1989
High	6603	0.2079	0.2933

Table 5.3.6.2 Boxplot by the absolute error of the price range

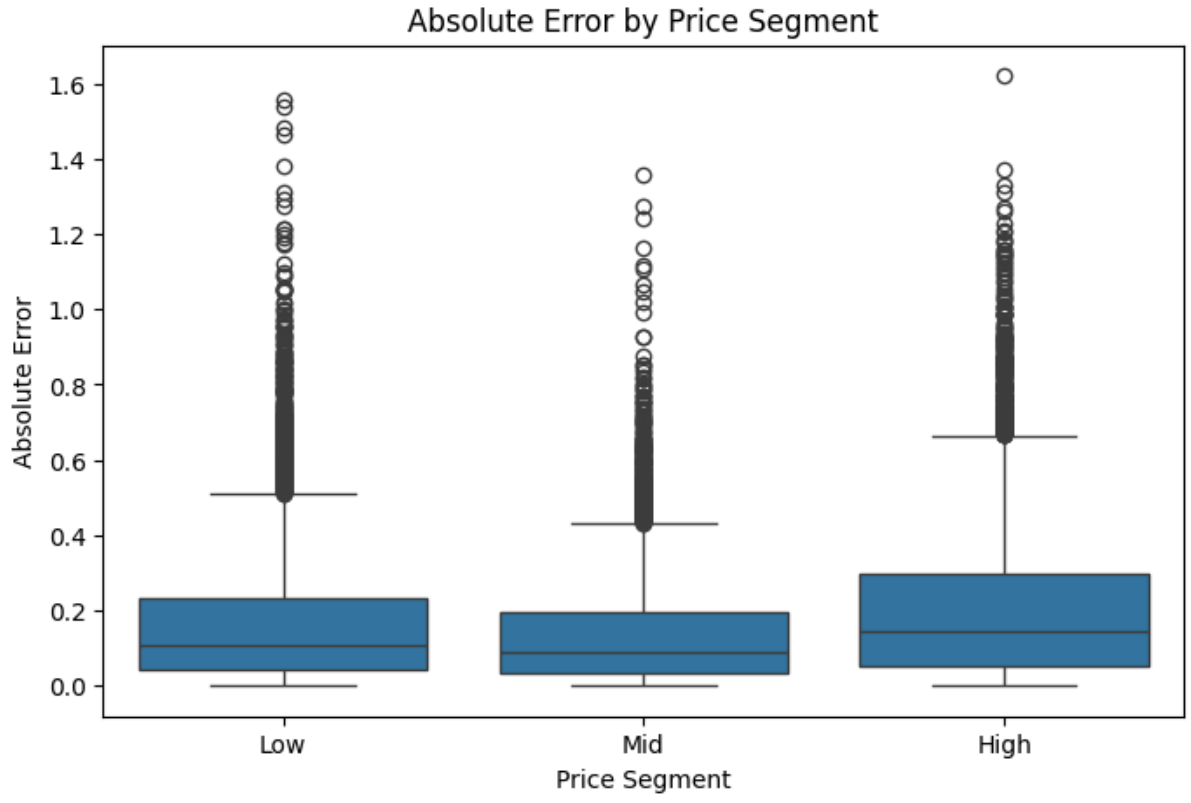


Table 5.3.6.3 Robustness table of different random seeds

Seed	MAE	RMSE
0	0.1674	0.2433

42	0.1721	0.2507
100	0.1685	0.2483

5.3.7 Stakeholder Meaning and Tradeoffs

The geographic contribution improves the practical value of the pricing framework by making rent estimates more sensitive to local market differences. For property managers and landlords, this means that pricing decisions can be based not only on broad state-level comparisons, but also on more localized spatial patterns. As a result, the model provides a more informative benchmark for identifying listings that may be overpriced or underpriced relative to similar units in nearby market areas.

This is also useful for renters. A pricing benchmark that includes both broad and localized geographic information can better distinguish whether a listing appears fairly priced or whether it may represent relatively better value within its local market context. In this sense, the geographic contribution supports more transparent and evidence-based comparisons across listings.

However, this improvement also involves tradeoffs. Compared with a simpler baseline using only structural features and broad geographic variables, the cluster-based geographic representation increases model complexity and reduces interpretability. While state-level effects are easy to explain directly, cluster-based features are less intuitive because they represent learned spatial groupings rather than clearly named real-world regions.

There is also a tradeoff between flexibility and simplicity. The refined geographic representation improves overall fair-value estimation when combined with state, but it is not strong enough to fully replace broader location information on its own. This means that the most useful pricing benchmark comes from combining interpretable regional features with more flexible localized spatial features, rather than relying on either type alone.

6. Discussion

6.1 Conclusion and Decision Interpretation

The project shows that apartment pricing can be estimated more effectively when the model includes more than structural housing features alone. In addition to unit size and room count, location and listing-level information help improve fair-value rent estimation.

For stakeholders, the main recommendation is to use model-based pricing benchmarks rather than relying only on simple comparisons or informal judgment. This can help landlords and property managers identify overpriced or underpriced listings more consistently, while also

giving renters a clearer sense of value. The practical impact is improved pricing transparency and more evidence-based rental decisions.

6.2 Limitations

The project has several limitations. First, some listing variables are incomplete or noisy, especially semi-structured fields such as amenities and pets. Second, the amenity and geographic modules were developed and evaluated separately, with different preprocessing and baseline choices, which limits full integration at the current stage. As a result, the findings are stronger within each module than as a single unified end-to-end framework. Third, the models are evaluated on historical listing data, so real-world deployment performance remains uncertain.

Because of these limitations, the model should be used as a pricing support tool rather than as a fully automatic decision system. It is more reliable for broad benchmarking than for replacing human judgment in unusual cases.

6.3 Future Work

Future development would focus on a hybrid strategy for further enhancement involving synthesizing feature engineering of amenity of group members with the coordinate-based geoclustering developed by other team members. While the current model relies on discrete administrative boundaries like City and State, utilizing spatial algorithms such as **K-Means** or **HDBSCAN** would allow the model to identify more organic pricing neighborhoods that transcend official borders. This hybrid approach, combining descriptive property features with hyper-local spatial context, represents the most direct route to minimizing remaining residuals and exceeding our current performance benchmarks.

While the current model is highly accurate, it demonstrates that geography is the primary driver of rental prices. Future work will focus on two primary frontiers: hyper-local feature engineering and advanced ensemble modeling. To push the R^2 beyond the 0.80 threshold, I intend to integrate **Geospatial APIs** to calculate proximity to public transit, high-performing schools, and major tourist attractions since we have the coordinates for all listings.

On the modeling side, the next step is to evaluate **CatBoost** as a replacement for XGBoost. Our current dataset relies heavily on categorical features (City, State, Amenities). While we used One-Hot Encoding (OHE) for this project, I learned that CatBoost is designed to handle categorical data natively using "Ordered Boosting”. This could potentially reduce the "dimensionality explosion" caused by OHE and better capture the relationship between specific cities and rental premiums without the risk of overfitting.

To push the performance even further, I want to implement a **Stacking Regressor**, which involves training multiple "base" models and then using a "meta-model" (like a simple Linear Regression) to learn how to best combine their predictions. Since different models have different blind spots, a stacked ensemble can learn that Model A is more accurate for luxury high-rises while Model B is better for suburban studios, leading to a more balanced final prediction.

By transitioning from a single-model approach to a meta-ensemble that incorporates precise distance-to-POI data, the project aims to further minimize predictive variance and provide a production-ready valuation tool for the modern rental market.

Member Contribution

Task	Member
Technical Report & Documentation	Brandon Lu / Xinyu Jing / Tri Nguyen / Wing Yan Kwan
Modeling w/ NLP-Driven Amenity Extraction & Categorical Feature Engineering (GitHub)	Brandon Lu
Modeling w/ Geospatial Feature Engineering & Geoclustering (GitHub)	Xinyu Jing
Presentation: Project Scope & Pipeline Architecture	Tri Nguyen
Presentation: NLP-Driven Amenity Methodology	Brandon Lu
Presentation: Spatial Intelligence & Market Cluster	Xinyu Jing
Presentation - Future Outlook, Conclusion	Wing Yan Kwan

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